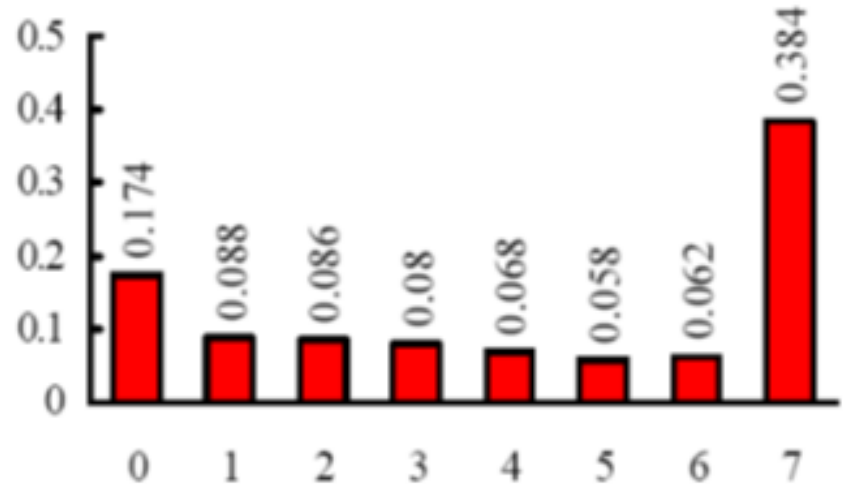
The histogram of image is as follow, please process the histogram equalization and outline the new gray level after equalization.



r_i	$p_r(r_i)$	$s_{k\text{计}}$	$s_{k\text{舍}}$	s_k	$p_r(s_k)$
$r_0=0$	0.174	0.174	1/7	$s_0=1/7$	0.174
$r_1=1/7$	0.088	0.262	2/7		
$r_2=2/7$	0.086	0.348	2/7	$s_1=2/7$	0.174
$r_3=3/7$	0.08	0.428	3/7		
$r_4=4/7$	0.068	0.496	3/7	$s_2=3/7$	0.148
$r_5=5/7$	0.058	0.554	4/7		
$r_6=6/7$	0.062	0.616	4/7	$s_3=4/7$	0.120
$r_7=1$	0.384	1	1	$s_4=1$	0.384

0、1、2、3、4、5、6、7 corresponding to 1、2、2、3、3、4、4、7 (2%)

For any point (x,y) in the image, the response, $g(x,y)$, of the filter is defined as

$$g(x,y)=\frac{1}{6}f(x-1,y)+\frac{1}{6}f(x,y-1)+\frac{1}{3}f(x,y)+\frac{1}{6}f(x,y+1)+\frac{1}{6}f(x+1,y)$$

(1) Give mask of this filter for mask or template operation.

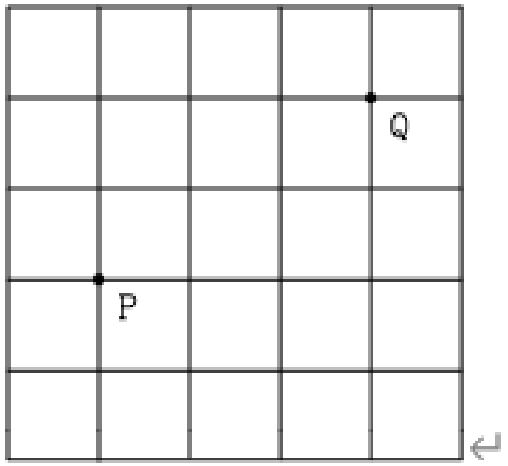
(2)Compute the result using this filter to the image shown as follow(Do not process the border of the image).

20	7	80	6
11	15	3	2
8	60	220	40
120	8	19	4

0	1/6	0
1/6	1/3	1/6
0	1/6	0

20	7	80	6
11			2
8			40
120	8	19	4

Considering the following figure, compute the Euclidean Distance, Chessboard distance and the City-block distance of the two points: P and Q.



5.

- Euclidean: $\sqrt{13}$, (2%)
  Chessboard:3, (2%)
  City-block:5 (2%)

There is an subimage which is the result of zooming. If we use nearest neighbor interpolation and bilinear interpolation respectively to get the value of new pixels, please give the values of the points of P1 and P2(both methods).

3			6
P1			
	P2		
4			7

Nearest neighbor interpolation:P1=3,P2=4; (4%)

bilinear interpolation:

$$P1=3\times\frac{2}{3}+4\times\frac{1}{3}=3\frac{1}{3};$$

$$P2=(3\times\frac{1}{3}+4\times\frac{2}{3})\times\frac{2}{3}+(6\times\frac{1}{3}+7\times\frac{2}{3})\times\frac{1}{3}=\frac{42}{9}$$ (6%)